

TERM PROJECT: COMBINED LOADING ANALYSIS & FEA VALIDATION

Objective: As Mechatronics Engineers, designing robotic links, structural frames, and mechanical components requires a deep understanding of how structures behave under complex loading conditions. The objective of this project is to evaluate a 2D or 3D body under combined loading. You will solve the problem analytically using classical mechanics and validate your findings numerically using Finite Element Analysis (FEA) software.

PROBLEM ASSIGNMENT & INITIAL ASSUMPTIONS:

You have been assigned a specific problem from the textbook. Please refer to the attached "**Project Assignment Sheet**" and find the problem associated with your Student ID number.

Before beginning your calculations, you must establish the following:

- **Material Selection:** You must choose a specific engineering material for your component (e.g., AISI 1020 Steel, 6061-T6 Aluminum, Gray Cast Iron). Clearly document the material name, whether it is **ductile or brittle**, and its relevant properties: Young's Modulus (E), Poisson's ratio (ν), and Yield (S_Y) or Ultimate (S_{UT}/S_{UC}) strength.
- **Assumed Dimensions:** If your assigned textbook problem is missing any cross-sectional properties (e.g., a specific diameter, wall thickness, or length), you must **choose reasonable engineering value**. Clearly state and box this assumed value at the beginning of your report and use it consistently across all analytical and computational steps.

PART 1: ANALYTICAL SOLUTION (HANDWRITTEN)

Note: This entire section must be solved by hand on paper using a pen/pencil. Calculations must be neat, logical, and easy to follow. Scans or clear photos of these pages will be included in your final report.

1. Static Analysis & Internal Loads

- Draw a complete, scaled Free Body Diagram (FBD) of the entire structure.
- Solve the global statics to find all reaction forces and moments at the supports.
- Determine the internal forces (N, V) and moments (M, T). Depending on your assigned problem, determine the critical cross-section where these loads are at an absolute maximum.
- Draw the **Shear Force and Bending Moment Diagrams** for the relevant axes of the body.

2. Stress Analysis

- Calculate the exact normal (σ) and shear (τ) stresses acting at the point of interest due to the combined axial, bending, shear, and torsional loads (incorporating your assumed dimensions where necessary).
- Draw a **3D Stress Element** representing the state of stress at that specific point.

3. Principal Stresses & Failure Theories

- Construct **Mohr's Circle** for your stress element to determine the Principal Stresses ($\sigma_1, \sigma_2, \sigma_3$) and the Maximum In-Plane/Out-of-Plane Shear Stress (τ_{max}).
- Based on the material you selected:
 - **If Ductile:** Calculate the equivalent stress using both the **Von Mises (Maximum Distortion Energy)** and **Tresca (Maximum Shear Stress)** criteria.
 - **If Brittle:** Calculate the failure stress using the **Maximum Normal Stress Theory**.

PART 2: COMPUTATIONAL ANALYSIS (FEA)

You must model your assigned problem using either **SolidWorks Simulation** or **ANSYS Static Structural**.

1. Setup & Pre-Processing

- Create an accurate 3D CAD model of your assigned body, ensuring it exactly matches your assumed dimensions.
- Apply the material properties for the specific material you selected in Part 1.
- Apply appropriate boundary conditions (fixtures/supports) and external loads.
- Generate an appropriate mesh. (Consider applying mesh controls/refinement around the critical point or cross-section).

2. Post-Processing & Results

Extract and capture clear screenshots of the following contour plots:

- Reaction forces at the supports (to verify statics).
- Normal and Shear stress distributions.
- Principal Stresses ($\sigma_1, \sigma_2, \sigma_3$).
- Von Mises Stress, Tresca Stress, OR Maximum Principal Stress (depending on your chosen material's ductility).
- Use the "Probe" tool to extract the exact stress values at the specific point analyzed in Part 1.

PART 3: DISCUSSION & COMPARISON

1. Data Comparison & Factor of Safety (FoS)

- Calculate the **Factor of Safety (FoS)** for your component based on your analytical calculations and your selected material's yield/ultimate strength.
- Calculate the **Factor of Safety (FoS)** based on the FEA results.
- Create a summary table comparing your Analytical Results vs. FEA Results for: Internal loads, Principal Stresses, Equivalent Stresses (Von Mises/Tresca/Max Normal), and FoS. Include a column for **Percentage Error**.

2. Technical Discussion

Write a 1-2 page discussion addressing the following:

- Did your analytical and numerical results match? Why or why not?
- Did your chosen material prove to be a safe choice for this application based on the calculated FoS? Would you recommend a different material or a change to your assumed dimensions based on the results?
- What assumptions are made in classical mechanics (e.g., Saint-Venant's Principle, idealized boundary conditions) that differ from how the FEA software calculates stress?
- Did you observe any stress concentrations in your FEA model that were not accounted for in your hand calculations?
- Discuss the impact of your mesh density on the accuracy of your FEA results.

DELIVERABLES & REPORT FORMAT

Submit a single, compiled PDF report with the following structure:

1. **Cover Page:** Project title, your name, student ID, and the assigned problem.
2. **Introduction & Assumptions:** Brief statement of the problem, a clear list of your selected material properties, and any assumed dimensions.
3. **Scanned Hand Calculations:** Clean, legible scans of all work for Part 1.
4. **FEA Report:** Screenshots of your mesh, boundary conditions, and all required stress/result plots with clear legends.
5. **Results Table & Discussion:** The comparative table, FoS calculations, and your technical discussion (typed).
6. **Conclusion:** A brief summary of the component's structural viability.

GRADING RUBRIC: COMBINED LOADING & FEA PROJECT

Part 1: Analytical Solution (40 Points)		
Criteria	Excellent (Full Credit)	Max Points
FBDs & Statics	FBD is perfectly scaled and labeled. Global statics and support reactions are 100% correct with clear logical steps.	10
Internal Loads & Diagrams	Shear and moment diagrams are completely accurate. Internal loads at the point of interest are correct.	10
Stress State & Mohr's Circle	Normal/shear stresses are correctly calculated. Stress element is accurate. Mohr's circle is drawn correctly to find principal stresses and the max shear stress.	10
Failure Theories	Yield/Failure criteria (Von Mises/Tresca/Max Normal) are appropriately selected and mathematically correct.	10

Part 2: Computational Analysis [FEA] (30 Points)		
Criteria	Excellent (Full Credit)	Max Points
Model Setup & Constraints	CAD model is flawless. Material properties, fixtures, and external loads perfectly match the analytical assumptions.	10
Mesh & Visual Plots	Mesh is appropriate (refined at critical zones). All requested contour plots (Reaction, Stress, Principal, Equivalent) are included and clearly legible.	10
Data Extraction	The "Probe" tool (or equivalent) is used correctly to extract exact stress values at the analytical point of interest.	10

Part 3: Discussion & Formatting (30 Points)		
Criteria	Excellent (Full Credit)	Max Points
Results Comparison & FoS	Summary table is clear and complete. Factor of Safety (FoS) and percentage errors between both methods are perfectly calculated.	10
Technical Discussion	Highly insightful discussion on why results differ (e.g., Saint-Venant's, mesh density, stress concentrations). Demonstrates deep mechanical understanding.	10
Assumptions & Professional Formatting	Material explicitly chosen (all properties listed). Assumed dimensions are clear and boxed. Report is compiled into a single PDF. Hand calculations are neat and scannable. Formatting is highly professional and easy to read.	10